
NHANESsample

A sample of data from the National Health and Nutrition Examination Survey (NHANES)

Description

Lead, blood pressure, and demographic variables from NHANES 1999-2018, downloaded from the nhanesA package. Data was filtered to adults 20 years of age or older with nonmissing blood lead level, blood pressure, and demographic information.

Usage

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Format

A data frame with 31,265 rows and 15 variables:

ID Respondent sequence number ("SEQN" in NHANES)

AGE Age ("RIDAGEYR" in NHANES: Best age in years of the sample person at time of HH screening. Individuals 85 and over are topcoded at 85 years of age up to 2006 and individuals 80 and over are topcoded at 80 years of age after 2006.)

SEX Gender ("RIAGENDR" in NHANES)

RACE Race and ethnicity ("RIDRETH1" in NHANES)

EDUCATION Education Level ("DMDDEDUC2" in NHANES: What is the highest grade or level of school you have completed or the highest degree you have received?)

INCOME Poverty income ratio (PIR): a ratio of family income to poverty threshold ("INDFMPIR" in NHANES)

SMOKE Smoking status (Combination of SMQ020 (Have you smoked at least 100 cigarettes in your entire life?) and SMQ040 (Do you now smoke cigarettes?) in NHANES: equal to "Still Smoke" if respondent answered "Yes" to SMQ020 and either "Every day" or "Some days" to SMQ040, equal to "Quit Smoke" if respondent answered "Yes" to SMQ020 and "Not at all" to SMQ040, and equal to "Never Smoke" otherwise.)

YEAR Year of the Study (Equal to the first year of the two year interval in which the response was recorded - NHANES surveys are grouped in two-year intervals)

LEAD Lead (ug/dL): "LBXBPB" in NHANES unless the reported level of lead was less than the lower limit of detection (llo), as defined by the paper cited above, for the relevant year, in which case "LBXBPB" was replaced by llo/sqrt(2))

BMI_CAT Body Mass Index Category (kg/m²): Based on "BMXBMI" in NHANES

LEAD_QUANTILE Quantile membership for blood lead levels based on the distribution of lead levels in the data

HYP Hypertension Status: Based on "BPQ020" (Have you ever been told by a doctor or other health professional that you had hypertension, also called high blood pressure?) and "BPQ040A" (Because of your high blood pressure/hypertension, have you ever been told to take prescribed medicine?) in NHANES. Equal to 1 if the respondent answered "Yes" to either of these questions, or, if data on either of these questions isn't answered, if SBP >= 130 or DBP >= 80, and equal to 0 otherwise.

ALC Alcohol Use: Based on "ALQ120Q" (In the past 12 months, how often did you drink any type of alcoholic beverage?) up to 2016 and "ALQ121" (the same question, but used after 2016) in NHANES. Equal to "Yes" if the respondent's answer to either of these questions was > 0 and equal to "No" otherwise.

DBP1 First Diastolic Blood Pressure (mmHg) reading: "BPXDI1" in NHANES.

DBP2 Second Diastolic Blood Pressure (mmHg) reading: "BPXDI2" in NHANES.

DBP3 Third Diastolic Blood Pressure (mmHg) reading: "BPXDI3" in NHANES.

DBP4 Fourth Diastolic Blood Pressure (mmHg) reading: "BPXDI4" in NHANES.

SBP1 First Systolic Blood Pressure (mmHg) reading: "BPXSY1" in NHANES.

SBP2 Second Systolic Blood Pressure (mmHg) reading: "BPXSY2" in NHANES.

SBP3 Third Systolic Blood Pressure (mmHg) reading: "BPXSY3" in NHANES.

SBP4 Fourth Systolic Blood Pressure (mmHg) reading: "BPXSY4" in NHANES.

Source

Data was obtained from the nhanesA package <https://CRAN.R-project.org/package=nhanesA>.

Variable selection and feature engineering were conducted in an effort to replicate the analyses conducted by

Huang, Z. (2022). Association Between Blood Lead Level With High Blood Pressure in US (NHANES 1999-2018). *Frontiers in Public Health*, 892.

<https://www.frontiersin.org/articles/10.3389/fpubh.2022.836357/full>.
